

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO5: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 30%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Analytical Problem-Solving Assessment

Duration: 1hr

1. A delivery robot operates in an uncertain environment. Apply the **SARSA** algorithm to update the action-value function and explain how the learned policy adapts to environmental changes. **(5 Marks)**

A. Problem Scenario

delivery robot must move from a starting location to a delivery destination while avoiding obstacles. The environment is uncertain because obstacles may move or appear unexpectedly.

The robot uses the SARSA algorithm to learn the best action in each situation.

1. Define the State

The state represents the robot's current situation.

S=(Robot Position, Obstacle Information)

For example:

State = Robot at (2,3)

Obstacle ahead = Yes

Target = (5,5)

2. Define the Actions

The robot can choose:

Action	Meaning
Up	Move forward/up
Down	Move backward/down
Left	Move left

Action	Meaning
Right	Move right

3. Define the Rewards

Situation	Reward
Reaches delivery target	+100
Normal movement	-1
Hits obstacle	-20
Invalid movement	-10

The robot learns to reach the target with fewer steps while avoiding obstacles.

4. Apply the SARSA Algorithm

SARSA stands for:

State → Action → Reward → State → Action

The SARSA update equation is:

$$Q(S,A) \leftarrow Q(S,A) + \alpha [R + \gamma Q(S',A') - Q(S,A)]$$

Where:

S = Current state

A = Current action

R = Reward received

S' = Next state

A' = Next action selected

α = Learning rate

γ = Discount factor

5. Analytical Example

Assume:

Current $Q(S,A) = 10$

Reward $R = -1$

Learning Rate $\alpha = 0.1$

Discount Factor $\gamma = 0.9$

Next $Q(S',A') = 15$

Using SARSA:

$$Q(S,A) = 10 + 0.1[-1 + 0.9(15) - 10] = 10 + 0.1[-1 + 13.5 - 10] = 10 + 0.1(2.5) = 10.25$$

Therefore:

Updated Q-value = 10.25

This means the selected action becomes more valuable based on the observed reward and the next action.

6. SARSA Learning Process

Start State



Select Action A



Perform Action



Observe Reward R



Observe Next State S'
↓
Select Next Action A'
↓
Update $Q(S,A)$
↓
Repeat Until Target Reached

7. Adaptation to Environmental Changes

Suppose the robot initially learns this path:

Start → Right → Right → Up → Target

Later, a new obstacle appears:

Start → Right → X Obstacle

The robot receives a negative reward:

Reward = -20

Using SARSA, the Q-value of moving right decreases.

The robot then explores alternative actions:

Start → Up → Up → Right → Target

If this new path produces better rewards, its Q-values increase.

Therefore, the learned policy changes from:

OLD POLICY:

Move Right

to:

NEW POLICY:

Move Up

8. Final Analysis

SARSA is an on-policy reinforcement learning algorithm. It updates the action-value function using the action that the robot actually chooses in the next state.

In an uncertain environment:

New obstacles produce negative rewards.

Unsafe actions receive lower Q-values.

Alternative safe actions are explored.

Successful paths receive higher Q-values.

The robot gradually adapts its navigation policy.

Conclusion

The delivery robot uses SARSA to continuously update its Q-values based on its actions and environmental feedback. When the environment changes, previously successful actions may receive negative rewards, causing their Q-values to decrease. The robot explores new actions and gradually learns an updated policy that selects safer and more efficient paths.

Code of Conduct:

I K.Mounith(192425090) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *K.Mounith*

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively